Exam 1 Review

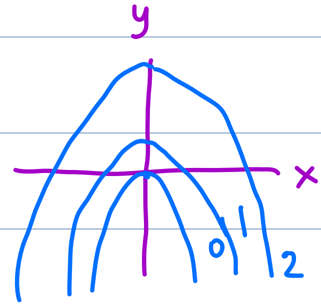
Surfaces

Example: Consider $x^2 - z = y^2$

What are the traces in:

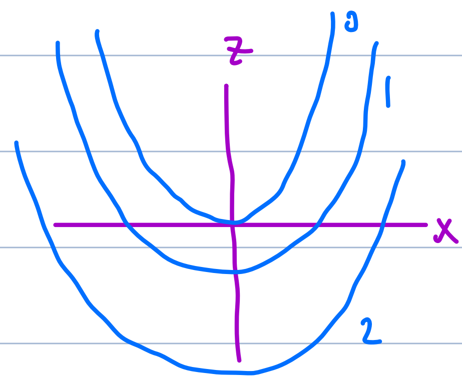
a) $x = k$ } $k^2 - z = y^2 \Rightarrow z = k^2 - y^2$

Parabola



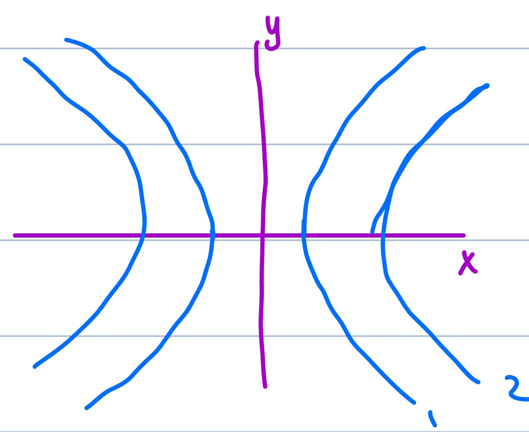
b) $y = k$ } $x^2 - z = k^2 \Rightarrow z = x^2 - k^2$

Parabola



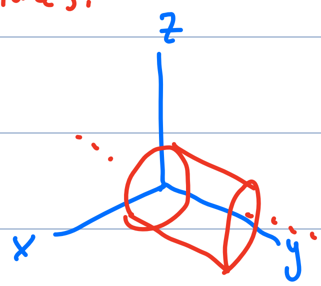
c) $z = k$ } $x^2 - k = y^2 \Rightarrow x^2 - y^2 = k$

hyperbola

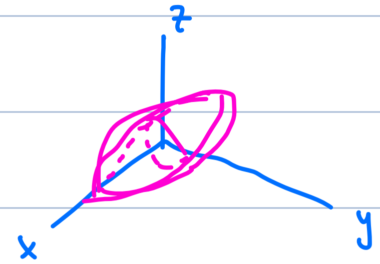


Question: identify the following surfaces:

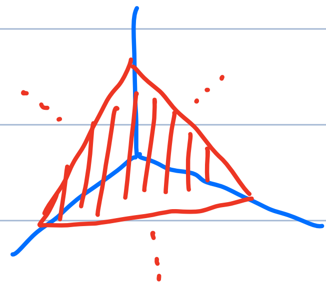
a) $x^2 + z^2 = 4$ cylinder



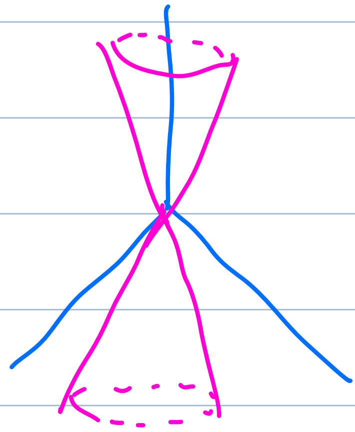
b) $x^2 + 2y^2 + 3z^2 = 4$ ellipsoid



c) $x + y + z = 4$ plane



d) $z^2 = x^2 + y^2$ cone



Question: is the following vector equation describing a line?

- $\vec{r}(t) = \langle 2+3t^2,\ 3-t^2,\ 4+t^2 \rangle$?

$$= \langle 2,3,4 \rangle + \langle 3t^2, -t^2, t^2 \rangle$$

$$= \underbrace{\langle 2,3,4 \rangle}_{\vec{r}_0} + \underbrace{t^2}\underbrace{\langle 3,-1,1 \rangle}_{\vec{v}}$$

yes, motion along a line
(always going along the direction $\langle 3,-1,1 \rangle$)

- $\vec{r}(t) = \langle 2+t,\ 3+t^2,\ 4-t \rangle$?

no

Curvature

Formulas: $K(t) = \dfrac{|\vec{T}'|}{|\vec{r}'|}$

$$K(t) = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}$$

Example: find the curvature of:

$\vec{r}(t) = \langle 2\cos(t), 2\sin(t), t \rangle$

$\vec{r}'(t) = \langle -2\sin(t), 2\cos(t), 1 \rangle$

$|\vec{r}'(t)| = \sqrt{4\sin^2(t) + 4\cos^2(t) + 1} = \sqrt{4(\underbrace{\sin^2(t)+\cos^2(t)}_{1})+1}$

$= \sqrt{4+1} = \boxed{\sqrt{5}}$

$\vec{T} = \dfrac{\vec{r}'(t)}{|\vec{r}'(t)|} = \dfrac{\langle -2\sin(t), 2\cos(t), 1\rangle}{\sqrt{5}} = \underbrace{\langle -\frac{2}{\sqrt{5}}\sin(t), \frac{2}{\sqrt{5}}\cos(t), \frac{1}{\sqrt{5}}\rangle}$

not annoying to differentiate

$\Rightarrow \quad \vec{T}' = \langle -\frac{2}{\sqrt{5}}\cos(t), -\frac{2}{\sqrt{5}}\sin(t), 0 \rangle = \dfrac{\langle -2\cos(t), -2\sin(t), 0\rangle}{\sqrt{5}}$

$|\vec{T}'| \overset{\star}{=} \dfrac{|\langle -2\cos(t), -2\sin(t), 0\rangle|}{\sqrt{5}} = \dfrac{2}{\sqrt{5}} = \dfrac{2}{\sqrt{5}}$

$K(t) = \dfrac{|\vec{T}'|}{|\vec{r}'|} = \dfrac{2/\sqrt{5}}{\sqrt{5}} = \boxed{\dfrac{2}{5}}$

$\star$ in general $|cv| = |c| \cdot |v|$

Intersections

Example: Find all intersection points where the curve

$$\vec{r}(t) = \langle \underbrace{t}_{x}, \underbrace{t^2}_{y}, \underbrace{3}_{z} \rangle$$

intersects the surface $x^2 + y = z^2$

$$\left.\begin{aligned} x &= t \\ y &= t^2 \\ z &= 3 \end{aligned}\right\} \xRightarrow{\text{plug into}} x^2 + y = z^2$$

$$\Downarrow$$

$$t^2 + t^2 = 9 \Rightarrow 2t^2 = 9$$

$$t^2 = \frac{9}{2} \Rightarrow t = \pm \frac{3}{\sqrt{2}}$$

Points: 1 $t = -\frac{3}{\sqrt{2}}$: $\vec{r}(t) = \langle -\frac{3}{\sqrt{2}}, \left(-\frac{3}{\sqrt{2}}\right)^2, 3 \rangle$

$$= \langle -\frac{3}{\sqrt{2}}, \frac{9}{2}, 3 \rangle \Rightarrow \left(-\frac{3}{\sqrt{2}}, \frac{9}{2}, 3\right)$$

2 $t = \frac{3}{\sqrt{2}}$: $\vec{r}(t) = \langle \frac{3}{\sqrt{2}}, \frac{9}{2}, 3 \rangle \Rightarrow \left(\frac{3}{\sqrt{2}}, \frac{9}{2}, 3\right)$

Equation of plane

Find the equation of the plane that contains the line of intersection :

$$\left[\begin{cases} x + y + z = 4 & ① \\ x + 2y + 3z = 6 & ② \end{cases}\right.$$

$\langle 1,1,1\rangle$ is perp. to line
$\langle 1,2,3\rangle$ is perp. to line $\Rightarrow$ $\langle 1,1,1\rangle \times \langle 1,2,3\rangle$ is parallel to line.

and is perpendicular to the plane $\underbrace{x - y - z = 3}_{\langle 1,-1,-1\rangle}$,

$\vec{n}$ perpendicular to $\langle 1,-1,-1\rangle$

Want: • point

set $z=0 \Rightarrow$ $\begin{cases} ① \; x + y = 4 \\ ② \; x + 2y = 6 \end{cases}$ ② − ① : $y = 2 \Rightarrow$ ① $x + 2 = 4$

$\Rightarrow x = 2$

point: $\langle 2,2,0\rangle$

• normal vector

$\langle 1,-1,-1\rangle$ is perpendicular to $\vec{n}$

$\Rightarrow \underbrace{\langle 1,-1,-1\rangle}_{\vec{u}}$ is parallel to our plane

To find normal, can find two parallel vectors and take their cross product.

To find another parallel vector:

• approach 1: find another point on line, take displacement

• approach 2: use cross product.

$$\begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = \vec{i}(3-2) + \vec{j}(1-3) + \vec{k}(2-1)$$

$$= \underbrace{\langle 1, -2, 1 \rangle}_{\vec{v}}$$

$$\vec{n} = \vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & -1 \\ 1 & -2 & 1 \end{vmatrix} = \vec{i}(-1-2) + \vec{j}(-1-1) + \vec{k}(-2+1)$$

$$= \langle -3, -2, -1 \rangle$$

$\vec{r}_0 = \langle 2, 2, 0 \rangle$ $\quad$ $\vec{n} = \langle -3, -2, -1 \rangle$

$$\vec{r} \cdot \vec{n} = \vec{r}_0 \cdot \vec{n}$$

$$\langle x, y, z \rangle \cdot \langle -3, -2, -1 \rangle = \langle 2, 2, 0 \rangle \cdot \langle -3, -2, -1 \rangle$$

$$-3x - 2y - z = -6 - 4 = -10$$

$$\boxed{3x + 2y + z = 10}$$

Curves

Example: Find parametric equations for the intersection of the paraboloid $z = x^2 + y^2$ ①

with the plane $\underline{x = 2}$ (2)

Try letting t represent one of the variables such that all the other variables can be written in terms of it.

(1) and (2) : $z = 4 + y^2 \Rightarrow z$ is a function of y

use y as parameter.

$t = y.$ $\Rightarrow z = 4 + y^2 = 4 + t^2$

$x = 2$

$$\left.\begin{array}{l} \\ \\ \\ \end{array}\right\} \begin{array}{l} x = 2 \\ y = t \\ z = 4 + t^2 \end{array}$$

$$\boxed{\vec{r}(t) = \langle 2, t, 4 + t^2 \rangle}$$

Arclength: Find arclength of $\vec{r}(t) = \langle 2\cos(t), 2\sin(t), t \rangle$ between $\underbrace{\langle 2, 0, 0 \rangle}_{t = 0}$ and $\underbrace{\langle -2, 0, \pi \rangle}_{t = \pi}$

$$L = \int_a^b |\vec{r}\,'(t)|\, dt = \int_0^{\pi} \sqrt{5}\, dt = \boxed{\sqrt{5}\,\pi}$$

$|\vec{r}\,'(t)| = \sqrt{5}$